

Each of the subitems below will be rated from 1-5:

- 1 = very positive aspect of this paper
- 2 = positive aspect of this paper
- 3 = neutral or irrelevant criterion for this paper
- 4 = negative aspect of this paper
- 5 = very negative aspect of this paper

YOU MUST ALSO PROVIDE COMMENTS TO THE AUTHORS IN PROSE. IT IS NOT ACCEPTABLE TO MERELY FILL OUT NUMBERS, AND RETURN THE REVIEW!

Paper Title: EchoStature: Spatial Demographic Conditioning for Pediatric Ejection Fraction Regression, with Subgroup-Stratified Evaluation

First Author Last Name: Narayan

Appropriateness for PSB

Is relevant to conference themes : 1

Is relevant to session themes : 1

Meets length restriction (12 pages): 1

Comment: The manuscript is well aligned with PSB’s emphasis on computational methods for clinically relevant biomedical problems, particularly through its focus on pediatric cardiac imaging, demographic conditioning, and subgroup-specific model evaluation. Its consideration of demographic heterogeneity also fits the session theme. The manuscript meets the 12-page length requirement.

Presentation

Is clearly written : 1

Appropriate title and abstract : 1

Adequate figures and tables : 2

Clear problem statement (inputs and outputs) : 2

Clear descriptions of algorithms : 2

Provides appropriate detail : 3

Describes scope and limitations of the approach : 1

Describes implementation status : 2

Adequate references : 2

Comment: The manuscript is clearly written, with the problem, model inputs, pipeline, outputs, and loss functions presented logically. The mathematical formulation is generally easy to follow, although some notation, such as EF\_hat, should be defined more clearly. The figures and tables effectively present the main findings, including performance variation across demographic subgroups. However, the synthetic demographic clips are not directly validated; their quality is inferred mainly from downstream EF regression performance. The authors should clarify the intended properties of these clips and how their quality is assessed. The ablation study is also incomplete, as it does not evaluate the contribution of the contrastive objective, a key component of the method. Ablations should also be conducted under consistent experimental settings. The limitations are well discussed, and the references are adequate.

Significance

Represents a biologically important advance : 3

Addresses a biologically important and realistic problem : 2

Represents a computationally important advance : 3

Addresses a computationally important problem : 2

Is likely to scale up to realistic problems : 3

Provides new evidence for an existing technique : 2

Comment: The manuscript addresses a clinically relevant problem by showing that pooled EF regression performance can mask substantial differences across pediatric age and body-size subgroups. The proposed synthetic demographic conditioning is interesting, but improvements in MAE, RMSE, and R-square are modest, so its advantage over conventional demographic conditioning remains limited. Statistical and clinical significance analyses would better establish the practical value of these gains. Additional robustness analyses for subgroup effects and validation on larger, more heterogeneous or external cohorts would also strengthen the evidence for generalizability.

Originality

Represents a novel approach or combination of approaches : 2

Has not been published before : 2

Points out differences from related research : 2

Reformulates a problem in an important way : 2

Comment: The manuscript presents a novel combination of rendered synthetic demographic conditioning and subgroup-stratified evaluation for pediatric EF regression. The use of synthetic clips to encode demographic attributes is an interesting departure from conventional vector-based conditioning, and the manuscript appropriately distinguishes the approach from related work. Although prior pediatric EF studies have reported age- and body-size-related performance differences, this work extends that line of investigation by combining subgroup-aware evaluation with a distinct synthetic demographic conditioning strategy.

Technical Content

Evaluates the effectiveness of implementation or techniques : 2

The approach is supported with sound arguments : 2

The approach is supported with theoretical analyses : 3

The approach is supported with experimental results : 2

The approach is technically sound : 2

Additional Comments for the Authors: The manuscript presents an interesting approach to demographic conditioning for pediatric EF regression and highlights the importance of subgroup-specific evaluation. The overall methodology appears technically sound and is supported by experimental results. However, several analyses would strengthen the technical evidence. In particular, the contribution of the contrastive objective should be evaluated through ablation, and ablation experiments should be conducted under consistent settings. The synthetic demographic clips should also be validated more directly, rather than relying only on downstream EF regression performance, to demonstrate that they capture the intended demographic information. Given the relatively

modest improvements in MAE, RMSE, and R-square, reporting statistical uncertainty or significance would help establish whether the gains are robust. Finally, validation on larger, more heterogeneous or external cohorts would further strengthen the generalizability of the findings.

Overall recommendation:

|               |          |        |
|---------------|----------|--------|
| <b>ACCEPT</b> | MARGINAL | REJECT |
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Your confidence in recommendation:

|        |               |      |
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| STRONG | <b>MEDIUM</b> | WEAK |
|--------|---------------|------|

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